Table 11.5 Estimated Market Impact Parameters

Scenario	a_1	a_2	a_3	a_4	b_1
All Data	708	0.55	0.71	0.50	0.98
SP500	687	0.70	0.72	0.45	0.98
R2000	702	0.47	0.69	0.55	0.97

** = Estimation Period: 1H2011.*

pressure from other investors. Quite often, however, market participants observe, and data confirms, that buy orders are less expensive to transact than sell orders. But we have found that this is due to managers selling stocks more aggressively, as well as survivorship bias where managers have a complementary stock to buy when prices become too expensive but do not have a complementary stock to sell when the company has fallen out of favor.

But what also occurs at this point in time is that the stock volatility has spiked and liquidity has decreased. Even in cases where there is more trading in the name (such as during the financial crisis), the amount of transactable liquidity is often much lower because there are several investors on the same side of the order as the manager, thus everyone is competing for the smaller liquidity pool. All of which increase the cost to trade. Portfolio managers buy in times of favorable conditions and sell in times of adverse conditions and market stress. For example, an index manager may hold 5% of the ADV of a stock in the portfolio. If this stock is suddenly deleted from the index the expected trading cost to liquidate the position will likely be greater than the expected cost of trading an order of 5% ADV. This is because all index managers who own the stock will have to sell the shares and liquidate the position from their portfolio and thus the aggregated market selling pressure will be equal to the aggregated number of shares that need to be sold. In many situations, the number of shares that need to be transacted due to an index reconstitution could be much greater than 100% of the stock's ADV. Thus, the trading cost for the index event trade is dramatically higher than the costs that would occur from a non-index event trade.

To further highlight this point we examined a $100 million small cap portfolio comprised of 100 stocks. The average order size corresponding to this investment amount is 35% ADV and the volatility 42%. If the portfolio is purchased via a full day VWAP strategy the expected cost is 106 bp. This is a very realistic cost estimate for a small cap strategy under normal market conditions. But now suppose that liquidity has dried